

The Hemic and Lymphatic System

Md. Mokbul Hossain, DVM, M, Sc, PhD, Post-doc.

Professor

Department of Pathology

Bangladesh Agricultural University

Mymensingh-2202

The Hemic and Lymphatic System

Composed of-

- Bone marrow
- Thymus
- Lymph nodes
- Spleen
- Bursa of Fabricius

Functions of bone marrow

Pluripotent stem cells give rise to all cellular elements of blood

- Production of erythrocytes called erythropoiesis
- Production of granulocytes called leukopoiesis
 - Neutrophils
 - Eosinophils
 - Basophils
- Production of monocytes called monocytopoiesis
- Production of platelets or thrombocytes called megakaryocytopoiesis

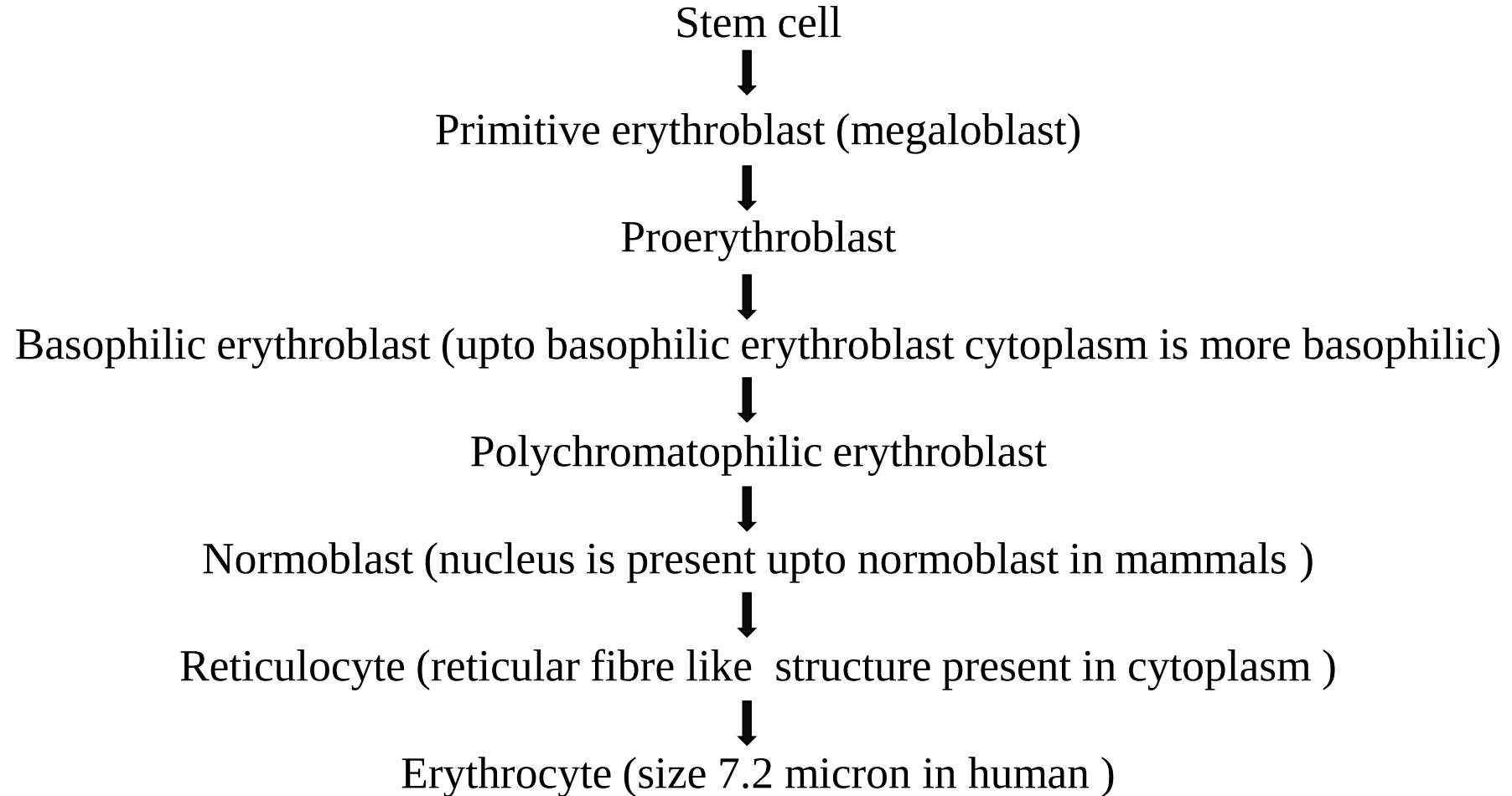
Functions of the Hemic System

- The hematic system is the body's **blood-producing and circulating network**.
- **Oxygen and Nutrient Transport:** Red blood cells carry oxygen to tissues, while plasma transports nutrients.
- **Hemostasis:** Platelets (thrombocytes) help clot blood to prevent excessive bleeding from injuries.
- **Waste Removal:** Blood transports metabolic waste to the kidneys and liver for excretion.
- **Immunity:** White blood cells (leukocytes) circulate in the blood to fight infections.

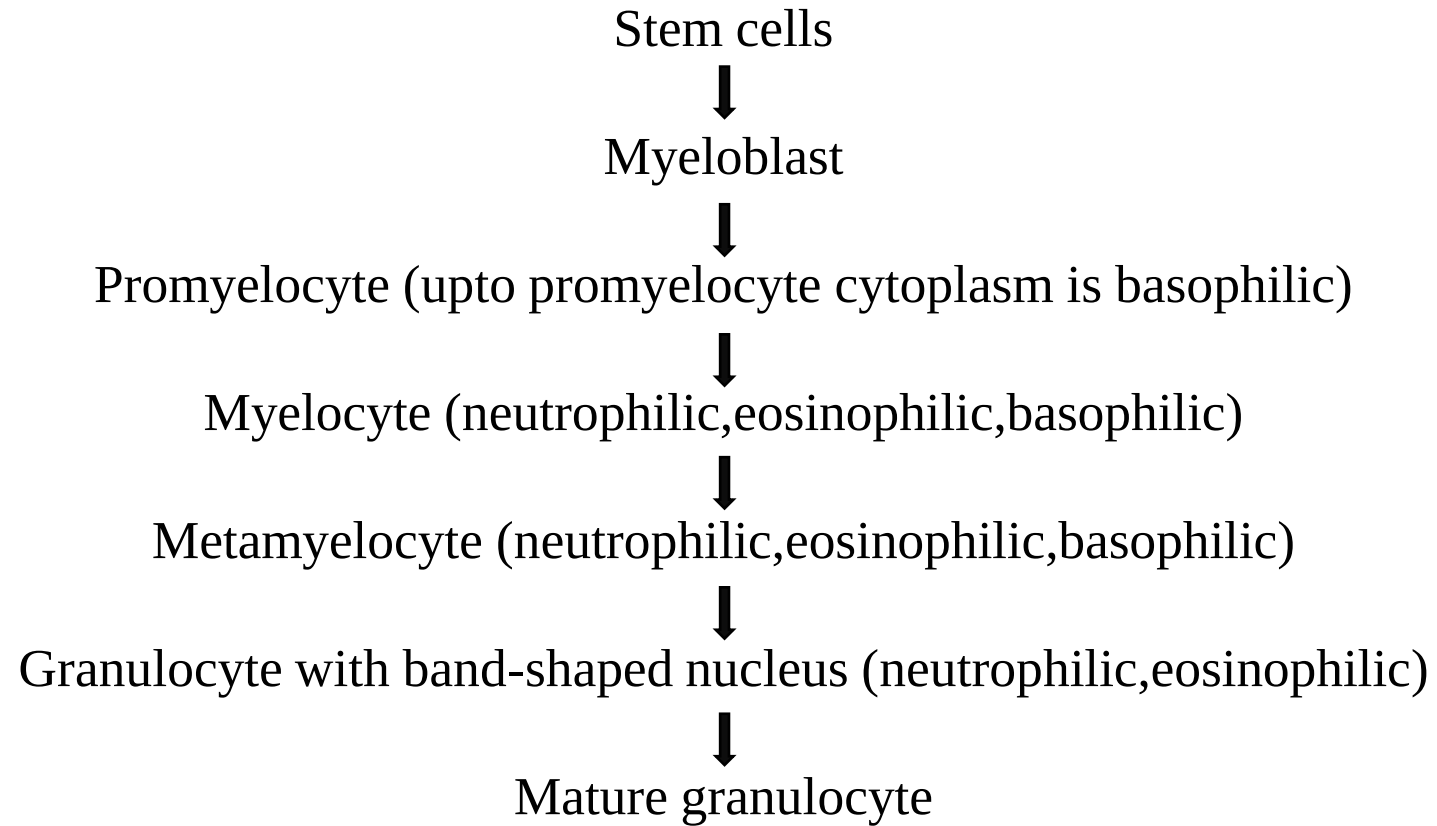
Functions of the Lymphatic System

- **Fluid Balance:** It collects excess fluid (lymph) that leaks from blood vessels into tissues and returns it to the bloodstream, preventing swelling (lymphedema).
- **Immune Defense:** Lymph nodes, the spleen, and the thymus filter lymph and blood, trapping and destroying pathogens (bacteria, viruses, and abnormal cells) while producing lymphocytes (white blood cells).
- **Fat Absorption:** Specialized lymphatic vessels in the small intestine (lacteals) absorb dietary fats and fat-soluble vitamins, transporting them into the bloodstream.

Erythropoiesis:



Leukopoiesis





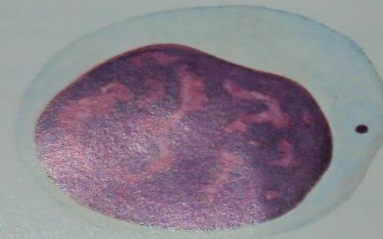
Neutrophilic granulocyte



Eosinophilic granulocyte



Basophilic granulocyte



Lymphocyte



Monocyte



Monocyte

The 5 Types of Human Leukocytes. (See Fig 13-5.)



Blastocyst



Blastocyst



Blastocyst with inner cell mass



Early embryonic development



Blastocyst



Early embryonic development



Early embryonic development



Early embryonic development



Early embryonic development



Early embryonic development



Blastocyst



Blastocyst with inner cell mass



Early embryonic development



Early embryonic development



Blastocyst



Blastocyst with inner cell mass



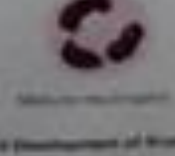
Early embryonic development



Early embryonic development



Blastocyst



Blastocyst with inner cell mass



Early embryonic development

Stages of Development of Embryos and Gametes. See Fig. 14.1.

- **Neutrophil** (small size granules, neutral color granules and usually 3 lobes of nucleus but it may be 3-5 lobes, size of mature cell is 12 micron in human)
- **Eosinophil** (large size granules, eosinophilic color granules and usually 2 lobes of nucleus, size of mature cell is 9 micron in human)
- **Basophil** (large size granules, basophilic color granules and usually s-shaped lobes of nucleus ,size of mature cell is 12 micron in human)

Monocytopoiesis

Monoblast

Monocyte (horseshoe-shaped nucleus, 9-12 micron)

Megakaryocytopoiesis

Megakaryoblast (ovoid/kidney-shaped nucleus with numerous nucleoli , size of the cell is 15-50 micron)

Megakaryocyte (a giant cell, size of cell is 35-150 micron, irregularly lobulated nucleus)

Thrombocytes/Platelets (2-5 micron in size, no nucleus but fragments of cytoplasm)

Lymphocytopoiesis

Lymphoblast

Prolymphocytes

Lymphocytes (round nucleus, scanty cytoplasm, 6-8 micron in diameter; originate from bone marrow and come to bursa of Fabricius of birds near cloaca but in mammals still is confused that they may come to fetal liver or developed in bone marrow then migrate to lymphoid organs producing HI. Some of lymphocytes come to thymus and they produce CMI)

In adult individual, bone marrow is the main organ for hematopoiesis. It includes

- Erythropoiesis
- Leukopoiesis

Embryologically, hematopoiesis occurs in

- Yolk sac
- Liver
- Spleen
- Lymphnodes
- Thymus

In rat and mice, extramedullary hematopoiesis occurs in

- liver
- Spleen
- Lymphnodes
- Adrenal gland
- Pituitary gland

Hematopoietic marrow is called red marrow. When individual grows to adult red marrow is replaced by fat marrow or yellow marrow and occupy most of the area of long bone. A little red marrow is found at

- Distal end of long bone
- A bit in the epiphysis

In addition to these two areas ,active red marrow is found in

- Flat bones
- Ribs
- Vertebrae

Why bone marrow impression smear is performed ?

- To know various cell types
- Various developmental stages
- Myeloid : erythroid ratio (3:1)

By bone marrow section, we can understand

- Mature granulocytes and their precursors and erythroid precursors (3:1)
- Aplasia of bone marrow
- Hypoplasia
- Hyperplasia
- Focal lesion e.g. granuloma
- Metastatic neoplasm

Biopsy is the best method to study the above lesions as bone marrow deteriorate rapidly. Bone marrow is collected by trocar and canula from the

- Rib
- Sternum
- Tuber coxae
- Iliac crest

Pluripotent stem cells undergo proliferation , development and maturation (hematopoiesis) and ultimately come to general circulation. But this hematopoiesis is controlled by some factors

- Erythropoietin from kidney
- Interleukins
- TNF-alpha
- GM-CSF (granulocyte-macrophage colony stimulating factors)
- Nutritional factors (Fe and vitamins)

Pathological conditions of bone marrow

A. Hyperplasia

Hyperplasia of bone marrow is grossly understood by increase in the amount of red marrow and decrease in the amount of fatty marrow. The color will be red but it requires the experience of normal for a given age.

Hyperplasia of bone marrow is called myeloid tissue hyperplasia and they are of two types

- a. **Erythroblastic hyperplasia** :grossly, the amount of red marrow is increased
- b. **Leukoblastic hyperplasia**: grossly, the grayish coloration of bone marrow is increased

a. Erythroblastic hyperplasia

Microscopically characterized by

- Increase in the no. of erythrocytes and normoblasts

Causes

- Most of the anemias except toxic aplastic anemia (due to inability of the bone marrow to function)
- Human pernicious anemia occurs due to deficiency of erythrocyte maturing factors
- If human and dog harbor a fish tapeworm, *Diphyllobothrium latum* the megaloblasts become numerous resulting megaloblastic anemia and megaloblastic hyperplasia.(very often erythroid hyperplasia is accompanied by myeloid hyperplasia)

b. Leukoblastic hyperplasia

Microscopically characterized by

- Increase in the no. of mature and immature neutrophilic granulocytes
- Increase in the no. of myelocytes and myeloblasts

Cause

- Pyogenic infection if acute then neutrophilic leukocytosis in blood and increase in no. of neutrophilic series of cells in bone marrow. But if chronic then neutrophilic leukopenia in blood and few myeloid cells in bone marrow due to exhaustion
- Ancylostomiasis and Trichinosis of human can cause eosinophilic leukocytosis in blood and increase no. of eosinophilic myeloblastic series of cells in bone marrow
- (Leukoblastic hyperplasia, malignant lymphoma and myeloid leukemia to bone marrow reveal “grayish coloration”

B. Hypoplasia

Less growth of bone marrow

Cause

- Toxic aplastic anemia
- Certain deficiency diseases

Gross

- Portion of fatty marrow is greater

Microscopic character

- Erythropoietic cells are less
- Less cellularity of all types of cells
- Fat cells are increased

Effect

- Anemia
- Decrease no. of all types of blood cells

C. Agranulocytosis

More or less complete absence of granulocytes from circulating blood due to complete aplasia of the leukocytic cells of the bone marrow

Cause

- Aplastic anemia
- Causes that suppress bone marrow

Effect

- Fatal as body can not protect infection

D. Osteomyelitis

Inflammatory process in the bone marrow due to infection.

Route of infection,

- Local wound e.g. fracture
- Hematogenous metastasis

Cause

- Pyogenic infection cause Purulent inflammation in bone marrow

Grossly

- characterized by painful accumulation of pus, softening and necrosis of overlying bony wall, and pus can make way to body surface.

Microscopically,

- it is composed of neutrophils, fibrins and other inflammatory cells
- TB cause granulomatous inflammation (Langhans type giant cell; acid-fast stain positive)
- Brucellosis cause granulomatous Inflammation (gram -negative rod)
- Coccidioidomycosis (*Coccidioides immitis*, different sized spores with endospores)

Osteomyelitis was difficult to treat but recently antibiotic has reduced some risks

E. Fibrosis

Fibrosis with or without myxomatous degeneration may occur in association with hypoplasia or aplasia

Cause

- Not known
- Exhaustion of active marrow may occur
- Myxomatous degeneration is prominent in starvation

Anemia

Reduction below normal the no. of erythrocytes and/or hemoglobin concentration or both per unit volume of blood.

Anemia is not a disease but manifestation of some other disease, e.g. Malaria can cause anemia

Rarely anemia is a primary disease, e.g. Fe and Cu deficiency can cause anemia

Anemia may be classified on the basis of

A. Morphology

B. Etiology

Morphological basis also depends on two features

Size of erythrocytes: MCV (Mean corpuscular volume; unit is cubic micron)

Hemoglobin content of erythrocytes: MCHC (Mean corpuscular hemoglobin concentration ; unit is %) and MCH (Mean corpuscular hemoglobin; unit is micrograms)

$$\text{MCV} = \frac{\text{PCV \% (42\%)}}{\text{Erythrocyte count in million /mm}^3 \text{ (6)}} \times 100 = 70 \text{ cubic micron}$$

$$\text{MCHC} = \frac{\text{Hb gm\% (14)}}{\text{PCV \% (42)}} \times 100 = 33.3 \%$$

$$\text{MCH} = \frac{\text{Hb gm \% (14)}}{\text{RBC million /mm}^3 \text{ (6)}} \times 10 = 23.3 \text{ microgram}$$

On the basis of MCV the size of erythrocyte may be

- **Macrocytic**
- **Normocytic**
- **Microcytic**

On the basis of MCHC the Hb content of the erythrocytes may be

- **Normochromic**
- **Hypochromic**
- **Hyperchromic:** More Hb can not be formed in RBC

Now the arrangement of the cell size and Hb content may be as follows

Macrocytic normochromic: Due to deficiency of vit. B12 ,folic acid, niacin

Macrocytic hypochromic: Due to acute blood loss, acute hemolysis

Normocytic normochromic : Toxic aplastic anemia (irradiation, bone marrow depressant agents)

Microcytic hypochromic: Deficiency of Fe, Cu, pyridoxine, chronic blood loss (cause Fe def.).Cu is essential for the utilization of Fe for the production of Hb.

Anemia is diagnosed by

- Clinical examination of Mm
- Enumeration /counting of erythrocytes
- Determination of Hb
- Calculation of MCV, MCHC

In addition to above mentioned tests,in a blood film or smear,the following changes are found

Anisocytosis: Erythrocytes are not uniform in sizes

Poikilocytosis : Bizarre or abnormal shaped erythrocytes

Nucleated erythrocytes : may be found in blood

Rubriblasts : may be found in blood

Basophilic stippling: dark specks are found in one cell

Polychromatophilia: Are not uniformly stained even in one cell or among the others

Heinz bodies: Purplish granules (denatured Hb) are found near the membrane of RBC

Target cell: Staining is found in the centre of the cell but the periphery is empty

Features in other tissues

Bone marrow: Red marrow is increased i.e. erythroblastic hyperplasia (except toxic aplastic anemia)

Liver and spleen : Regain hematopoietic function in severe and prolonged anemia that is called "extramedullary hematopoiesis". In rodents, extramedullary hematopoiesis is normal.

Gross appearance

- Mucous membrane and other tissues are pale
- At autopsy/necropsy, blood is pale and watery
- Grayish dots indicative of hematopoietic centres in other tissues (liver, spleen etc.) that may be confused with necrosis

Hyperplastic bone marrow : Grossly red and microscopically increase in No. of erythrocytic series of cells

On the basis of etiology anemia is classified into five groups

1. Hemorrhagic anemia
2. Hemolytic anemia
3. Deficiency anemia
4. Toxic aplastic anemia
5. Myelophthisic anemia

1. Hemorrhagic anemia

Results from severe hemorrhages.

Hemorrhage is divided into two types

- a) **Acute hemorrhage** that cause acute hemorrhagic anemia, hemorrhagic shock and death
- b) **Chronic hemorrhage** can cause chronic hemorrhagic anemia

Cause of acute hemorrhagic anemia

- Trauma
- Thrombocytopenic purpura
- Sweet clover and warfarin poisoning
- Bracken fern poisoning
- Trichloroethylene-extracted soybean oil meal

Chronic hemorrhagic anemia : Continued loss of blood through small hemorrhages

- Unhealed peptic ulcer of Human
- Hook worm
 - *Bunostomum phlebotomum* : cattle
 - *Ancylostoma duodenale* :human
 - *A. caninum* : Dog. All are found in Intestine.
- Stomach worms
 - Haemonchus spp (Barber's Pole Worm, Large Stomach Worm, Wire Worm) in Cattle
 - Ostertagia ostertagi (Medium or Brown Stomach Worm) in Cattle
 - Trichostrongylus axei (Small Stomach Worm) in Cattle
 - Mecistocirrus digitatus in Cattle
 - Strongylus vulgaris* : Horse. The adult worm of all three species are found in cecum and colon. Larvae are found in anterior mesenteric artery.
 - S. equinus* : Do
 - S. edentatus* :Do
- Hemophilia A & B:
 - In hemophilia A : Def. of Factor VIII.
 - In hemophilia B Def. Factor IX/Christmas factor
- Vit. C deficiency

Morphologically acute hemorrhagic anemia may be normocytic normochromic followed by normocytic hypochromic and macrocytic hypochromic. (4-6 weeks required for completion of regeneration of blood cells). If Fe stores approaches exhaustion then microcytic hypochromic anemia develops

2. Hemolytic anemia

Hemolysis of circulating erythrocytes are diagnosed by

- Hemoglobinemia : If intravascular hemolysis then
- hemoglobinemia and hemoglobinuria occur.
- Hemoglobinuria
- Hemolytic jaundice
- Stimulation of bone marrow in chronic cases resulting nucleated RBC and reticulocytes in circulation

Cause

- a) Infectious
- b) Toxic
- c) Immunogenic

a) Infectious

- Piroplasmosis
- Anaplasmosis
- Eperythrozoonosis of swine and cattle
- Hemobartonellosis of dog and cat
- Malaria of human, monkey and birds
- Equine infectious anemia
- Trypanosomiasis
- *Streptococcus hemolyticus*
- *Clostridium hemolyticum bovis*
- *Cl. welchii*
- Leptospirosis

b) Toxic causes

- Snake venoms
- Pot. and sodium chlorate poisoning
- Chronic lead poisoning
- Chronic copper poisoning
- Ricin (toxic principle of castor bean)
- Phenylhydrazine
- Saponin
- poisoning by kale and rape

c) Immunologic causes

- **Erythroblastosis fetalis** : due to Rh factor
- **Autoimmune hemolytic anemia** : failure to recognize the self

3. Deficiency anemia

Deficiency of several compounds and vitamins can cause anemia

Def. of Fe which is essential for Hb synthesis

Morphologically, it is microcytic hypochromic anemia

- Def. in diet (mothers milk or food)
- defective assimilation
- Malabsorption
- Chronic blood loss

Def. of copper : Required for utilization of Fe for Hb synthesis. Excessive molybdenum inhibit Cu absorption and ultimately interference of Hb synthesis. The type of anemia is microcytic hypochromic

Dietary def. of cobalt lead to def. of vitamin B12 which is required for development and maturation of erythrocytes. The type of anemia is macrocytic normochromic.

Dietary def. of vitamin B12 in monogastric animal and man. Killing of rumen flora by oral antibiotic or some abnormal food through the cobalt supply is ok

Folic acid def. : Required for development and maturation of erythrocytes lead to macrocytic normochromic anemia

Pyridoxine def. : Lead to microcytic hypochromic and sideroblastic anemia. Some cells show Prussian-blue positive materials in erythrocytic cells of bone marrow.

Riboflavin, Vit. C/ascorbic acid, Vit. E and Selenium deficiency : Lead to anemia. Vitamin C cause scurvy and anemia in man, nonhuman primates and guineapigs. Selenium and Vit.E give the stability of plasma membrane of RBC and if deficiency occurs then destruction of RBC and anemia develop.

Pernicious anemia

Erythrocyte maturing factor (EMF) is stored in liver.

EMF has two components

- a) Intrinsic factor and
- b) Extrinsic factor i.e. vit B12

a) Intrinsic factor : a glycoprotein secreted by parietal cells. Deficiency of intrinsic factor may occur due to

- atrophic gastric gland (achylia gastrica) and
- Absence of HCl (achlorhydria).

b) Extrinsic factor i.e. vit B12 --Def. of vit B12 may occur due to

- Diarrhoea and
- dietary lack of animal protein (Vitamin B12 is naturally bound to the proteins in animal foods. In the stomach, digestive acids and enzymes strip the vitamin away from these proteins, allowing it to bind to "intrinsic factor" so it can be fully absorbed into your body)

Glycoprotein and vit B12 form a complex called EMF absorbed through ileum and stored in liver

Sometimes liver fail to store it due to cirrhosis

If man harboring a tapeworm *Diphyllobothrium latum* cause impair assimilation of Vit B12. In dog anemia occurs due to hemolytic toxin. (Fish muscle contain proceroid and plerocercoid and eaten by man and dog and the worm become adult in the intestine)

4. Toxic aplastic anemia

Aplasia means no growth. Bone marrow is injured in such a way, there is no production of erythroblastic and myeloblastic series of cells.

Gross

- Bone marrow is predominantly fatty
- scattered islands of hematopoietic cells
- sometimes sclerotic

Causes

- Toxic irradiation e.g. X-ray, radium, isotopes significant doses are lethal
- Benzol poisoning, a classical example
- Trinitrotoluene
- Sulphonamides (if individual is hypersensitized)
- Chloramphenicol
- Bracken fern
- Trichloroethylene-extracted soybean meal

Other causes

- **Nephritis and uremia**
 - toxic action of retained waste product
 - Interference of erythropoietin production.
- **Chronic infection:** has effects upon Hb synthesis
- **Neoplastic disease** : selective depression of erythrocytogenesis

5. Myelophthisic anemia

(Myelo, the marrow; phthisis, a wasting disease)

This type of anemia results from physical destruction of erythropoietic tissue, usually from its replacement by metastatic tumor tissues. (Metastatic tumor can cause atrophy, necrosis and ultimately anemia)

Causes

- Various carcinomas
- Leukemia
- Malignant lymphoma
- Osteopetrosis
- Infection e.g TB, Brucellosis, Coccidioidomycosis

Significance and effect of anemia

- Anoxia of the tissues
- Anoxia may lead to fatty change in heart, liver (paracentral necrosis) and other susceptible organs.
- Rapid and irregular pulse
- Dyspnea/shortness of breath
- Muscular weakness
- Hemolytic anemia may lead to
- Hemolytic jaundice
 - Hemosiderin in RE cells of spleen, bone marrow, liver etc. (Excess hemobilirubin in blood and tissue, excess urobilinogen in urine, more stercobilin and urobilin in feces and hemoglobinuria).
- Hemorrhage due to anoxic damage of endothelia
- Edema due to
 - Hypoxia lead to increased permeability
 - Protein loss in hemorrhagic anemia
 - Helminthiasis leads to drainage of RBC resulting decreased osmotic pressure.

Removal of cause can cure anemia except toxic aplastic anemia

Another classification of anemia

1. Hereditary anemia : Genetically determined defects in erythropoiesis or in viability of RBC result in anemia.

Example

- Tail short anemia of the mouse
- Flex-tailed anemia
- Spherocytosis of the mouse etc.

2. Immuno-hemolytic anemia

There will be ag and ab reaction causing lysis of RBC and development of anemia

Example

a) Isoimmune hemolytic anemia/erythroblastosis fetalis

Occurs due to Rh factor which is present on the surface of RBC. If father is Rh-positive and mother is Rh-negative then the child may be Rh-positive. The Rh factor of child can enter to mother blood and production of anti-Rh factor occurs. The anti-Rh antibody (first time IgM and second time IgG) can pass the placenta to fetus and cause lysis of RBC.

For the first time the effect is not so severe and in the next pregnancy the child may die even in the womb of mother. The anti-Rh or like factor can pass through placenta in human, rabbit and nonhuman primates and through colostrum in horse, swine and dog. The synonyms are neonatal isoerythrolysis, icterus neonatorum, neonatal jaundice

b) Idiopathic immune hemolytic anemia/autoimmune hemolytic anemia: RBC is coated with autoantibody directed against them, become spherical in shape (spherocytes; spherocytosis) and easily destroyed by RE system of the body especially spleen. The cause mainly idiopathic but may be associated with following conditions

Causes of idiopathic hemolytic anemia

- Neoplastic disease (lymphoma)
- Viral infections
- By drugs (para-amino salicylic acid, phenacetin , penicillin)

Autoimmune hemolytic anemia in human is determined by Coombs antiglobulin test

In human autoimmune hemolytic anemia is divided into three categories.

Warm antibody type : IgG antibody causing erythrocyte agglutination at 37° C

Cold antibody type : IgM antibody causing erythrocyte agglutination at 0-4° C

Cold hemolysin type : IgG antibody bind to erythrocytes at cold temperature and causes hemolysis 30°C

Autoimmune hemolytic anemia is not common in animals but SLE-like syndrome has been reported in dog, horse and cat.

RBC is coated with autoantibody directed against them, become spherical in shape (spherocytes ; spherocytosis) and easily destroyed by RE system of the body especially spleen.

Other lesions are

- active erythropoiesis with thrombocytopenia
- Splenomegaly
- Lymphadenopathy
- Purpura
- Glomerulonephritis
- lameness may be found due to rheumatoid arthritis

Blood grouping

Blood grouping are made by the presence of Ag on the surface of RBC such as agglutininogen, hemolysinogen which interact with corresponding antibody producing agglutination and/or hemolysis of the red blood cells.

In human ABO and Rh systems are used.

Group	Erythrocyte (Ag)	Serum (Ab)	Rh (+/-)
A	A	B	
B	B	A	
AB	A,B	-(universal recipient)	
O	-	A,B (universal donor)	

Rh factor / RhD Ag (Rhesus D antigen): Anti-D immunoglobulin (e.g., RhoGAM), which neutralizes fetal red blood cells before her body can react.

Rhesus monkey contains a protein known as Rh protein on the surface of erythrocytes. If antibody is raised against Rhesus monkey erythrocytes and this antibody reacts (agglutination and or hemolysis) with human erythrocytes then that human, is Rh-positive.

In animals different blood groups are known, but for transfusion of blood only agglutination and hemolysis are determined. Blood from donors is mixed with serum from recipient if no reaction, then the blood is used for trasfusion.

Some terminology

A. Polycythemia : Increase in no. of erythrocytes in circulation.

Three types

1. Relative polycythemia: Relative increase in No. of erythrocytes

Cause

- Vomition
- Diarrhoea
- Hemorrhage

2. Absolute polycythemia/ Erythrocytosis:

Absolute increase in No. of erythrocytes

Causes

- Prolonged mild anoxemia,
- High altitude
- Sufficient anoxia
 - Patent foramen ovale
 - Pulmonary emphysema
 - Pulmonary fibrosis

3. Polycythemia vera : Absolute increase in No. of erythrocytes but cause is not known

B. Reticuloendotheliosis :

Myeloproliferative disease of cat characterized by proliferation of undifferentiated reticulum cells in bone marrow, spleen (splenomegaly), liver (hepatomegaly), lymph nodes (enlarged) and these cells can enter into circulation.

Cause

- Not known. Vit B 12 and folic acid deficiency mimic the disease. The role of feline leukemia virus is not yet established.

C. Purpura hemorrhagica :

Common in horse but also occurs in swine. Denotes a syndrome characterized by many petechial and ecchymotic hemorrhages in the skin and external and internal mucous membrane. Latin word purpura means purple colored hemorrhagic spots. The condition occurs at recovery stage of some diseases and it is thought that the injury of the capillaries occurs due to hypersensitivity or immune complex deposition.

In this purpura hemorrhagica, no thrombocytopenia and nor coagulation defect has been determined.

Cause

- Strangles (*Streptococcus equi*)
- Other respiratory infection

D. Thrombocytopenic purpura: A reduction in no. Of circulating platelets results purpura hemorrhagica owing to prolongation of bleeding time

Causes

- Destructive lesions of the bone marrow
 - Lymphomatous and other neoplastic invasion
 - Ionizing radiation
 - Causes of aplastic anemia (bracken fern poisoning, trichloroethylene extracted soybean meal)
- Exhaustion of supply of platelets in repeated attempts to seal capillary leakage
- Equine infectious anemia
- Tick-borne fever in sheep and cattle by *Rickettsia phagocytophilia*
- Idiopathic thrombocytopenia
- Immunologic thrombocytopenia : common in baby pigs. The mechanism is similar to rh-factor. Factor of platelets of baby pig sensitize mother and antibody is produced against platelets factor. Destruction of baby pigs platelets after having clostrum.

Petechial and ecchymotic hemorrhages in skin, internal and external mucous membrane and subcutaneous tissue. Megaloblast is absent in bone marrow.

Diseases of leukocytes

❑ **Leukocytosis** : Inflammatory increase in no. of leukocytes determined by TLC (20000-30000/mm³)

Cause

Pyogenic infection leads to neutrophilia determined by DLC. Immature neutrophil is called juvenile/stab/band (2-6%) and counted by schilling count.

Schilling count is divided into two

- Shift to the left (immature neutrophils increase) and
- Shift to the right (mature neutrophils increase)

❑ **Leukopenia** : decrease in no. of leukocytes in circulation. e.g. Rinderpest , bovine viral diarrhea , infectious bursal disease (IBD), HIV, FIV,SIV,BIV etc

❑ **Leukemia** : neoplastic increase in no. of leukocytes in blood where involvement of bone marrow is very common.

❑ **Hemoglobinemia and hemoglobinuria:**

Hb releases after hemolysis of erythrocytes producing hemoglobinemia and if exceed kidney threshold, then come out through urine called hemoglobinuria that must be differentiated from hematuria. After sedimentation or centrifugation erythrocytes settle down at the bottom of the test tube. The effects are hemosiderosis in RE cells, hemolytic anemia and hemolytic jaundice.

Causes are the causes of hemolytic anemia.

❑ **Post parturient hemoglobinuria :**

Occurs in cows that have given birth to calves a few weeks previously.

Clinical pathology and pathology:

- Severe hemolysis of circulating erythrocytes
- Hemoglobinemia
- Hemoglobinuria
- Active hematopoiesis i.e nucleated RBC in circulation
- Centrilobular necrosis and fatty change(due to anoxia)
- Erythrocytes contain "heinz bodies"
- Hemolytic jaundice
- Edema
- Hemoglobinuric nephropathy

Cause

- **Hypophosphatemia** : the membrane of RBC becomes instable due to deficiency of phosphorous causing hemolysis
- Cu deficiency

❑ **Heinz bodies anemia**

Heinz bodies are present in RBC that are denatured Hb due to some toxic hemolytic anemia. The toxic substances are rape, kale, onions, phenothiazine etc.

Mechanism

Oxidative injury to erythrocytes and production of glutathione and hydrogen peroxide and ultimately oxidation of Hb producing denatured precipitates in RBC called Heinz bodies.

Myoglobinuria:

Not related with hemic and lymphatic system but this must be differentiated from hemoglobinuria. Myoglobin size is 1/4 of size of Hb. So myoglobin (20mg/100ml in blood) has high renal clearance and Hb (100 mg/100 ml in blood) has low renal clearance.

Spectroscopic technique is needed to differentiate between hemoglobinuria and myoglobinuria

Four genetic diseases of leukocytes

1. Chediak-Higashi syndrome

Man, cattle, cat

clinical sign

- Dwarf
- Photophobia (albinism)
- hemorrhagic tendency
- susceptible to infection
- Giant granules are found in neutrophils, eosinophils, basophils lymphocytes, hepatocytes, tubular epithelia, neurons, endothelial cells and melanocytes.

cause is genetic defect.

2.Cyclic neutropenia (gray Collie Syndrome)

Disease is characterized by a periodic neutropenia occurring at regular intervals of 8-12 days. Neutrophils may completely disappear from the peripheral blood at 10.5 to 11.5 day intervals, only to reappear with a “rebound” neutrophilia. Bone marrow transplantation from healthy animal to affected animal relieve the clinical sign.

Affected bone marrow to healthy animal develop cyclic neutropenia.

Clinical signs are

- cyclic neutropenia and gray hair,
- diarrhoea,
- lameness and
- chronic ulceration in oral and genital mucosa.

3. Pelger-Huet anomaly :

- Man, rabbit, and dog.
- Nuclei of maximum neutrophils and eosinophils are nonsegmented .

4. Leukocyte adhesion deficiency :

A hereditary disease recorded in Holstein-Friesian calves, dog and -human.

Cause :

- Deficiency or absence of beta 2 integrins or adhesion to endothelium prior to their extravasation as well as their role in phagocytosis
- Neutrophilia occurs but can not go out blood vessels

Spleen

A great reticulo-endothelial sponge and composed of red pulp and white pulp/Malpighian follicle. B lymphocytes are found in centre and T lymphocytes surround this B zone.

Function

- Filtration of blood to remove foreign elements and old erythrocytes ("graveyard" for old erythrocytes)
- Serve as a reserve pool of erythrocytes
- An organ of immune system, a source of lymphocytes
- A site for hematopoiesis under appropriate condition

Splenic enlargement, splenomegaly

Different mechanisms can cause swelling of the spleen

- Circulatory disturbances
- Inflammatory disease
- Metabolic disease
- Neoplastic disease

a) Acute congestion of the spleen

Etiology

- Acute myocardial failure
- Euthanasia by Nembutal or similar barbiturates
- Torsion can cause hemorrhagic infarct

Gross

- Enlarged
- Somewhat soft
- Cut surface bulges and oozing of blood occurs

Microscopic

- A large no. of erythrocytes in the sinuses and cords of Billroth

b) Chronic passive congestion of the spleen

Gross

- Enlarged
- Firm/little tough due to fibrous tissue proliferation

Microscopic

- Distension of sinusoids and cords with blood
- Hyperplasia of sinusoidal endothelia becoming cuboidal
- Diffuse fibrosis through out the pulp
- Thickening of trabeculae
- Hemosiderin in phagocytes

Etiology

- Cirrhosis of the liver
- Congestive heart failure or right-sided heart failure causing back pressure on splenic vein

c) Hematomas (cyst of hemorrhage) of the spleen in dog

Nodular enlargement, fibrous capsule around it and may adhere to omentum.

Microscopically characterized by blood or a mass of lysed blood, and degenerating granulocytes

d) Infectious cause of splenomegaly

Infectious may be acute and chronic

Acute splenitis

Anthrax

- congestion
- influx of neutrophils
- Increase in No. of endothelial cells, MQs and lymphocytes
- Severely unclotted blood-distended spleen of acute septicemia called "blackberry-jam spleen"

Chronic splenitis

- Brucellosis
- Toxoplasmosis
- Gaseous lymphadenitis
- Pseudotuberculosis
- Leishmaniasis

Many of the diseases may lead to lymphoid hyperplasia

e) **Storage disease and deposition disease**

Example

Gaucher's disease : Lipid-laden RE cells in spleen due to deficiency of glucocerebrosidase

Amyloidosis : Sago spleen (amyloid deposition around central artery) and becon spleen(amyloid deposition along with reticular fibre)

f) **Anemia may lead to splenomegaly**

Two different mechanisms may cause enlargement of spleen /splenomegaly.

Extravascular hemolysis : No. of MQs increases resulting splenomegaly.

Many chronic anemia may lead to extramedullary hematopoiesis

(Thrombocytopenic purpura may lead to subcutaneous hemorrhage and splenic enlargement)

g) **Proliferative and neoplastic disease**

- Lymphosarcoma
- Hemangioendothelioma
- Fibrous histiocyoma
- Metastatic neoplasms etc

Hyposplenism

- small spleen
- Death following acute hemorrhage
- Hereditary immunodeficiency e.g SCID foal, SCID mice where spleen, thymus and lymphnodes are small.
- Aplasia of the spleen (rare).

Lymphnodes

Composed of lymphoid follicles which have B lymphocytes in the centre and T lymphocytes at the periphery (paracortical region). Presentation of ag occurs in dendritic cells and MQs. Antigenic stimulation to lymphnodes can increase in No. B and T lymphocytes depending on the nature of Ag. If antigenic stimulation then the No. of plasma cells comparatively increases. Immunologic stimulation must be differentiated from follicular lymphoma.

Lymphadenitis

Lymph nodes are not glands but the inflammation of the lymph nodes is called lymphadenitis, which is a misnomer however, a commonly used terminology.

Causes : Two

Specific:

- Anthrax : (*Bacillus anthracis* ; Hemorrhagic lymphadenitis)
- Strangles : (*Streptococcus equi* ; abscess formation in horse, puppies)
- Mycotic infection : e.g. Histoplasmosis
- Immunosuppression by HIV, SIV, FIV
- TB: -*Mycobacterium bovis*
 -M. *Tuberculosis*
- Pseudotuberculosis : *Yersinia pseudotuberculosis*
- *Corynebacterium pseudotuberculosis*
- Ovine caseous lymphadenitis
- Tularemia: *Francisella tularensis*
- Actinobacillosis : *Actinobacillus lignieresii*

Nonspecific

- Pneumonia affects bronchial lymph nodes (hyperemia, edema)
- Rhinitis or an infected tooth affects mandibular and pharyngeal lymph nodes
- Mastitis affects supramammary lymph nodes (The nonspecific lymphadenitis is called lymph nodes draining reaction)

Other lesions of lymphnodes

- Hypoplasia: e.g. SCID mice, foal or other animals
- Anthracosis
- Fluke infection : Black pigments are found
- Hemosiderosis
- Hemochromatosis
- Melanosis
- Porphyrin pigment (brownish discoloration) in bovine lymph nodes
- Hyalin is found in germinal centers

Thymus

Function

T cell production and maturation and this thymus undergoes involution with the age

Anatomy

Immature T cells are found in the cortex and mature T cells are found in the medulla

Hassall's corpuscles are found in medulla composed of onion layers of keratinized epithelial cells

Some pathological conditions

Hypoplasia e.g. SCID mice

Atrophy : Immunodeficiency (HIV, SIV , FIV etc)

Hyperplasia : B cells lymphoid follicles bearing germinal centers in the medulla and compressing the cortex.

Cause is autoimmune response to thymic myoid cells .(similar to s/m)

Neoplasms

a) Thymoma: arises from epithelia and common in old animal.

b) Thymic lymphoma : common in young

Neoplastic diseases of hematopoietic and lymphatic tissues

Leukocytosis: 20000- 30000/mm³

Tumor

1. **Leukemia:** >30000/mm³

Originate from blood, stem cells

Involvement of bone marrow

Leukemia is of two types:

A. a. Leukemic leukemia

b. Aleukemic leukemia

Histology:

- Erythrocytic series of cells
- Myelocytic (N,E,B) series of cells
- Monocytic cells
- Megakaryocytic cells
- Mast cells

B. a. Acute leukemia: more than 30% blastic cells of ANC of bone marrow

b. Chronic leukemia: <30% blastic cells of ANC of bone marrow

2. Lymphoma:

A. Anatomical classification

- Multicentric
- Thymic
- Alimentary
- Splenic
- cutaneous

B. Histological classification

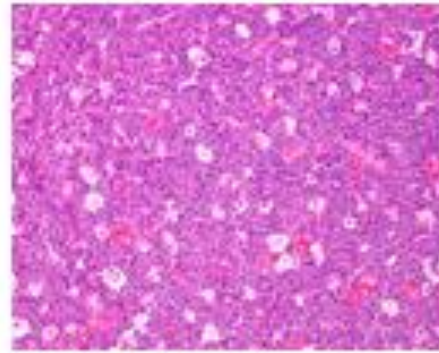
- B cell lymphoma
- T cell lymphoma
- Blastic lymphoma
- Lymphocytic lymphoma
- Histiocytic

3. Myeloma

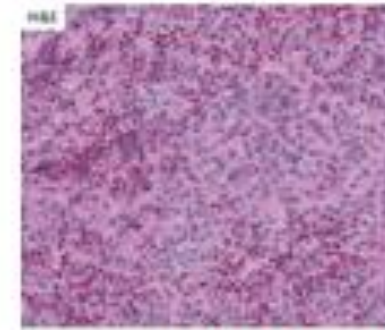
- Plasma cell tumor

4. Burkitt's lymphoma:

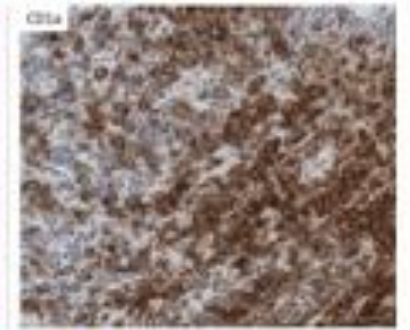
- Medium cells
- Blastic cells
- Histiocytes
- Starry sky pattern with macrophages



Burkitts lymphoma

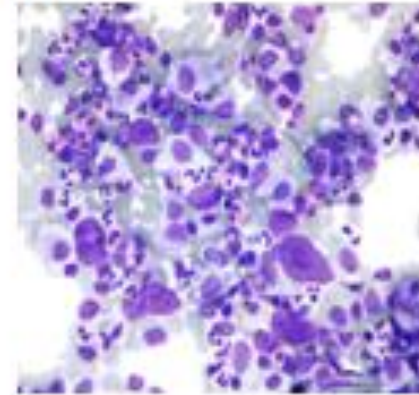


Histiocytic lymphoma

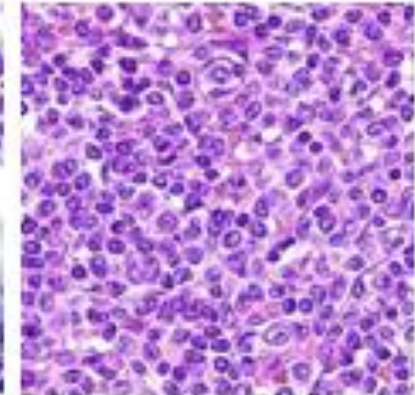


5. Hodgkin's disease:

- Lymph nodes are affected
- Plasma cells
- Eosinophils
- Fibroblast
- Reed Sternberg giant cells(owl eyed)



Hodgkin lymphoma



Non Hodgkin lymphoma

6. Histiocytic lymphoma

The Epstein-Barr Virus (EBV) causes Hodgkin's lymphoma. ...Non-Hodgkin lymphoma. ...Burkitt's lymphoma. ...